

Negative-Ray Summation of Countable Exponential-Feedback Transseries

A sharp slope threshold, an inverse Bessel representation,
and logarithmic-Gevrey error certificates

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Abstract

Consider the formally well-defined feedback equation

$$\widehat{U}(q) = \sum_{j \geq 1} q^j \exp(\lambda_j \widehat{U}(q)), \quad \lambda_j \geq 0,$$

and its compositional inverse $\widehat{Q}$. We prove that $\widehat{U}$ and $\widehat{Q}$ are *finely Borel–Laplace summable in the negative direction* if and only if $\lambda_j = O((j \log(j+1))^2)$. The sums are inverse functions and recover the literal feedback equation near the negative real axis. The proof inverts the feedback equation before taking its Borel transform: an exact signed exponential-block expansion becomes a normally convergent sum of Bessel kernels. If $A = \limsup \lambda_j / (j \log j)^2 < \infty$, this constructs the inverse Borel transform on the parabolic region $|\zeta| + \operatorname{Re} \zeta < 1/(2A)$; for $A = 0$ it constructs an entire function. For the quadratic model $\lambda_j = j^2$, we additionally prove a uniform remainder estimate on a closed left half-disc. Its explicit positive majorant M_N satisfies

$$\log M_N = N \log N - 2N \log \log N + (\log 4 - 1)N + o(N).$$

Truncation at $N \sim \log^2(1/r)/(4r)$ consequently gives an error at most $\exp[-(1/4 - o(1)) \log^2(1/r)/r]$. We provide exact rational numerical certificates, an inverse-error transport bound, reproducible programs, and a research agenda separating fine summability from angular summability, resurgence, and actual optimal-error asymptotics.

Status. The theorems below are conventional mathematical proofs, not Lean-verified statements. The proposed new contribution is the constructive negative-direction summation theorem and its quantitative remainder theory for this model. The underlying Lagrange, Bessel, and nonlinear summation tools are classical. Global publication priority has not been established. In particular, we do not claim a general transseries summation theorem or a solution of a named longstanding conjecture.

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1 The repository question and the result obtained

1.1 A precise boundary between existing work and this article

The ProveIt transseries project studies formal asymptotic scales, well-based supports, differential blocks, inversion, and analytic error transport. At the snapshot used here, the material has its own directory `Analysis/Transseries/`. The exact snapshot is

a795fcffa4b3ece22761d81fbed3c43be8b81795.

The principal point of departure is the research package *A sharp regularity classification for countable exponential-feedback transseries* [2]. Its README explicitly distinguishes the proved left-sector Poincaré realization from a uniform Gevrey remainder theorem and from directional Borel summation. Those latter assertions were not established there. This article addresses that specific boundary.

The earlier package already proves the formal existence and Gevrey classification of the feedback solution. In particular, its classification implies that the coefficient series is Gevrey-1 precisely at the growth threshold $(j \log j)^2$. We do not present that coefficient classification as a discovery of the present paper. Our main result is that, for this nonnegative-slope model, the same threshold is exactly sufficient for *fine negative-direction Borel summation*, with identification of the summed function. We give an independent proof of the necessity needed here, so our main theorem does not rely on accepting the earlier draft's asymptotic proofs without review.

The focused source review covered the current project README, the series-and-transseries directory README, and the feedback article and its README. It was not an exhaustive claim-by-claim audit of the much larger canonical transseries volume. The directory README also marks the recent research arrivals as unmerged and not formalized [1]. The formalization roadmap in Section 12 is accordingly prospective.

1.2 Why inversion first changes the analytic problem

Directly expanding $\widehat{U}$ produces nonnegative, potentially very large coefficients. However, its inverse is characterized by

$$u = \Phi(Q(u), u), \quad \Phi(q, u) = \sum_{j \geq 1} q^j e^{\lambda_j u}. \quad (1.1)$$

For $\operatorname{Re} u \leq 0$, every coefficient $e^{\lambda_j u}$ has modulus at most one. The countably infinite kernel is therefore controlled by a geometric series in q , regardless of how large the slopes are. Inverting in q while keeping u fixed retains that information. Expanding everything as an ordinary power series at the outset discards it.

The resulting exact inverse has blocks of the form $u^k e^{bu}$. Their integrated Borel transforms are Bessel kernels. On the negative Borel ray, positive b produces oscillation instead of exponential growth; the few possible negative frequencies satisfy a linear bound in k . This is the mechanism behind the theorem. It is a structural property of the defining equation, not a general consequence of Gevrey regularity.

1.3 Contribution map

The new analytic assertions are the threshold equivalence of Theorem 2.3, the explicit inverse Borel representation and parabolic continuation of Theorem 2.4, and the quadratic remainder and error-certificate results of Theorems 2.5–2.6. The proof also gives an all-slope continuous ray transform, even when no Borel germ exists. Classical tools are proved in the specialized form

needed here and credited at their first use. Numerical checks supplement, but do not establish, the infinite-dimensional or asymptotic conclusions.

2 Conventions and main theorems

Write

$$\widehat{U}(q) = \sum_{n \geq 1} u_n q^n, \quad \widehat{Q}(u) = \sum_{n \geq 1} v_n u^n, \quad \widehat{Q} = \widehat{U}^{(-1)}. \quad (2.1)$$

The formal inverse exists because $u_1 = 1$. All slopes are finite real numbers, and are nonnegative throughout. Natural logarithms are used. A statement about $j \log j$ starts at $j = 2$.

Definition 2.1 (Gevrey-1 and integrated Borel transform). A formal series $\widehat{f}(u) = \sum_{n \geq 0} f_n u^n$ is in $\mathcal{G}_1$ when $|f_n| \leq CH^n n!$ for some $C, H > 0$. Its integrated Borel transform is

$$\mathcal{B}_0 \widehat{f}(\zeta) = \sum_{n \geq 0} \frac{f_n}{n!} \zeta^n. \quad (2.2)$$

Thus $\widehat{f} \in \mathcal{G}_1$ exactly when (2.2) is a holomorphic germ at zero. The usual Borel minor of a series with zero constant term is the derivative of (2.2).

Definition 2.2 (Fine summability on the negative ray). Let

$$\Omega_\delta^- = \{\zeta \in \mathbb{C} : \text{dist}(\zeta, (-\infty, 0]) < \delta\}.$$

A series is *finely Borel summable in direction π* when its Borel germ extends to some Ω_δ^- and, there, satisfies $|\mathcal{B}_0 \widehat{f}(\zeta)| \leq C e^{H|\zeta|}$. Its sum at $u = -r$, for sufficiently small $r > 0$, is

$$\mathcal{S}_\pi \widehat{f}(-r) = \frac{1}{r} \int_0^\infty e^{-s/r} \mathcal{B}_0 \widehat{f}(-s) ds. \quad (2.3)$$

For complex u the same formula is $-u^{-1} \int_0^\infty e^{s/u} \mathcal{B}_0 \widehat{f}(-s) ds$, on $\text{Re}(-1/u) > H$.

This is the fine, half-strip version of summability, as distinguished from summability in an open arc of directions in [5, Sections 7 and 9]. Using the Borel minor gives an equivalent definition: integration preserves exponential growth, and differentiation preserves it after narrowing the strip. The domain in the u -plane is a disc tangent to the origin from the left, not a sector of opening greater than π .

Theorem 2.3 (Sharp negative-direction threshold). *For every sequence $\lambda_j \geq 0$, the following five conditions are equivalent:*

- (i) $\lambda_j = O((j \log(j+1))^2)$;
- (ii) $\widehat{U} \in \mathcal{G}_1$;
- (iii) $\widehat{Q} \in \mathcal{G}_1$;
- (iv) $\widehat{U}$ is finely Borel summable in direction π ;
- (v) $\widehat{Q}$ is finely Borel summable in direction π .

When they hold, the summed functions are compositional inverses near the negative real axis. In particular, for small negative q ,

$$\mathcal{S}_\pi \widehat{U}(q) = \sum_{j \geq 1} q^j \exp\left(\lambda_j \mathcal{S}_\pi \widehat{U}(q)\right), \quad (2.4)$$

and the sum on the right converges absolutely. The analogous identity holds on a sufficiently small common complex neighborhood contained in the left sectors where the dependent variable has negative real part.

Theorem 2.4 (Constructive inverse continuation). *Set*

$$A = \limsup_{j \rightarrow \infty} \frac{\lambda_j}{(j \log j)^2}. \quad (2.5)$$

If $0 < A < \infty$, the explicit kernel series of Section 5 defines $\mathcal{B}_0 \hat{Q}$ holomorphically on

$$\mathcal{P}_A = \left\{ \zeta \in \mathbb{C} : |\zeta| + \operatorname{Re} \zeta < \frac{1}{2A} \right\}. \quad (2.6)$$

It has exponential growth on a sufficiently thin fixed tubular neighborhood of the negative ray. If $A = 0$, the same kernel series is entire and still has this negative-ray growth property. The set (2.6) is a guaranteed continuation region, not a claim that its boundary consists of singularities.

The quadratic case has a stronger uniform conclusion. It is convenient to retain the convergent factor e^{-u} exactly and write

$$P(u) = e^u Q(u), \quad \hat{P}(u) = \sum_{n \geq 1} a_n u^n. \quad (2.7)$$

Let $\rho = 3 - 2\sqrt{2}$ and let C_k be defined in (4.4) below.

Theorem 2.5 (Uniform logarithmic-Gevrey remainder). *Suppose $\lambda_j = j^2$. Boundary values refer to the continuous block-series extension constructed below. For $0 < r_0 < \rho$, $N \geq 2$, and $\operatorname{Re} u \leq 0$, $|u| = r \leq r_0$, one has*

$$\left| P(u) - \sum_{n=1}^{N-1} a_n u^n \right| \leq r^N M_N(r_0), \quad (2.8)$$

where the entirely explicit positive majorant is

$$M_N(r_0) = \frac{\rho^{-N}}{1 - r_0/\rho} + \sum_{k=1}^{N-1} C_k \frac{[k(k-1)]^{N-k}}{(N-k)!}. \quad (2.9)$$

For fixed r_0 ,

$$\log M_N(r_0) = N \log N - 2N \log \log N + (\log 4 - 1)N + o(N). \quad (2.10)$$

The error for approximating $Q(u)$ by $e^{-u} \sum_{n < N} a_n u^n$ is at most e^r times the right side of (2.8).

Theorem 2.6 (A certified beyond-all-orders scale). *In the quadratic case, put $L = \log(1/r)$ and $N(r) = \lfloor L^2/(4r) \rfloor$. For every $\varepsilon > 0$ and all sufficiently small $r > 0$,*

$$\sup_{\substack{|u|=r \\ \operatorname{Re} u \leq 0}} \left| P(u) - \sum_{n < N(r)} a_n u^n \right| \leq \exp \left[- \left(\frac{1}{4} - \varepsilon \right) \frac{L^2}{r} \right]. \quad (2.11)$$

The same conclusion holds for the exponentially normalized approximation to Q , after possibly decreasing the admissible r .

Here and below, “optimal” in a theorem label refers to the leading-order optimization of our proved majorant. It does not assert a matching lower bound for the actual truncation error. Cancellation can make that error smaller. No Stokes constant is inferred from the number $1/4$.

3 Formal existence and the necessary growth condition

3.1 Local finiteness and positive Lagrange weights

Although Φ need not be analytic near $(0, 0)$, it is legitimate formally: its coefficient of q^n uses only $j \leq n$. If $V, W \in q\mathbb{R}[[q]]$ agree through degree $n-1$, then $\Phi(q, V)$ and $\Phi(q, W)$ agree through degree n . Iteration starting at zero therefore converges coefficientwise to a unique fixed point. Its first coefficient is one, and every coefficient is nonnegative.

For completeness, classical Lagrange inversion [3] yields a particularly useful finite coefficient formula. Introduce an auxiliary parameter t in $V = t\Phi(q, V)$, invert with respect to t , and then set $t = 1$ coefficientwise. Each coefficient of q permits only finitely many powers of t . For a finitely supported sequence of nonnegative integers $\mathbf{m} = (m_1, m_2, \dots)$, set $|\mathbf{m}| = \sum m_j$ and $w(\mathbf{m}) = \sum jm_j$. The result is

$$u_n = \sum_{w(\mathbf{m})=n} \frac{(\sum_j \lambda_j m_j)^{|\mathbf{m}|-1}}{\prod_j m_j!}. \quad (3.1)$$

The convention $0^0 = 1$ handles single-component terms. This identity is only a finite formal calculation; it imposes no analytic condition on the sequence of slopes.

Taking one component of size $j \geq 2$ and m components of size one in (3.1) gives

$$u_{j+m} \geq \frac{(\lambda_j + m\lambda_1)^m}{m!} \geq \frac{\lambda_j^m}{m!}. \quad (3.2)$$

This elementary inequality already detects the correct necessary scale.

Lemma 3.1 (Necessity of the logarithmic-square threshold). *If $\widehat{U} \in \mathcal{G}_1$, then $\lambda_j = O((j \log(j+1))^2)$.*

Proof. Choose $C, H \geq 1$ with $u_n \leq CH^n n!$, and put $m = \lceil j \log(j+1) \rceil$. From (3.2),

$$\lambda_j \leq C^{1/m} H^{1+j/m} (m!(j+m)!)^{1/m}.$$

Using $n! \leq n^n$, the last factor is at most $m(j+m)^{1+j/m}$. Now $j/m = O(1/\log j)$, so $(j+m)^{j/m}$ stays bounded, while m and $j+m$ are both $O(j \log(j+1))$. The remaining factors are bounded as well. The finitely many small j are absorbed into the constant. $\square$

3.2 Why reversion does not change class membership

Lemma 3.2 (Gevrey-1 reversion). *Let $f(z) = z + \sum_{n \geq 2} f_n z^n$ and let $g = f^{(-1)}$ be formal. Then $f \in \mathcal{G}_1$ if and only if $g \in \mathcal{G}_1$.*

Proof. It suffices to prove one direction, since $f = g^{(-1)}$. Lagrange's coefficient formula is

$$g_n = \frac{1}{n} [z^{n-1}] \left(1 + \sum_{j \geq 1} f_{j+1} z^j \right)^{-n}.$$

Expand the negative power. A term with ℓ factors is indexed by a composition $j_1 + \dots + j_\ell = n-1$, with $j_i \geq 1$, and has a binomial factor $\binom{n+\ell-1}{\ell}$. If $|f_m| \leq CH^m m!$, the product of factorials in that term obeys

$$\prod_{i=1}^{\ell} (j_i + 1)! \leq 2^{\ell-1} (n - \ell + 1)! \leq 2^n n!. \quad (3.3)$$

To see the first inequality, move a unit from the smaller of two parts greater than one to the larger; the factorial product can only increase. Iterating leaves one large part and $\ell - 1$ parts equal to one.

The binomial factor is at most 4^n , the number of all relevant compositions is at most 2^n , and the factors involving C and H are at most D^n for a fixed D . Summing gives $|g_n| \leq E^n n!$ after enlarging E . This proves class membership, not preservation of the sharp factorial-normalized type. $\square$

The preceding lemma is included to make the logic of necessity independent of any sharp asymptotic classification. Fine summability implies Gevrey-1 by local analyticity of the Borel germ. Thus either fine summability assertion in Theorem 2.3 already forces the slope condition. The rest of the paper proves its sufficiency constructively.

4 An exact signed exponential-block inverse

4.1 The coefficient measures

Fix u temporarily and invert the map $q \mapsto \Phi(q, u)$ at zero, with an independent target variable w . Lagrange inversion gives

$$q(w; u) = \sum_{k \geq 1} \frac{w^k}{k} [q^{k-1}] \left(\sum_{j \geq 1} q^{j-1} e^{\lambda_j u} \right)^{-k}.$$

Now set $w = u$. For $k \geq 2$, let $1 \leq \ell \leq k - 1$ and let $\mathbf{m} = (m_1, \dots, m_\ell)$ range over ordered positive compositions of $k - 1$. Define

$$\omega_{k,\ell} = \frac{(-1)^\ell}{k} \binom{k + \ell - 1}{\ell}, \quad (4.1)$$

$$b_{k,\mathbf{m}} = \sum_{i=1}^{\ell} \lambda_{m_i+1} - (k + \ell) \lambda_1. \quad (4.2)$$

Let μ_k be the finite signed atomic measure placing mass $\omega_{k,\ell}$ at $b_{k,\mathbf{m}}$ for each indexed composition. Coincident atoms are added. Put $\mu_1 = \delta_{-\lambda_1}$. Formally,

$$Q(u) = \sum_{k \geq 1} u^k \int_{\mathbb{R}} e^{bu} d\mu_k(b). \quad (4.3)$$

Every ordinary Taylor coefficient uses only $k \leq n$, so (4.3) also defines a formal series without convergence hypotheses. It solves $u = \Phi(Q(u), u)$ formally. Uniqueness of the formal feedback fixed point after substitution $q = Q(u)$ gives $\widehat{U}(Q(u)) = u$, identifying this series with the compositional inverse in (2.1).

4.2 A universal combinatorial majorant

Define $C_1 = 1$ and, for $k \geq 2$,

$$C_k = \frac{1}{k} \sum_{\ell=1}^{k-1} \binom{k + \ell - 1}{\ell} \binom{k - 2}{\ell - 1}. \quad (4.4)$$

There are $\binom{k-2}{\ell-1}$ relevant compositions, so $\|\mu_k\|_{\text{TV}} \leq C_k$.

The first seven values are 1, 1, 3, 11, 45, 197, 903.

Lemma 4.1 (Generating function and usable bounds). *The generating function of these positive majorants is*

$$S(t) = \sum_{k \geq 1} C_k t^k = \frac{1 + t - \sqrt{1 - 6t + t^2}}{4}. \quad (4.5)$$

Its radius of convergence is $\rho = 3 - 2\sqrt{2}$, and

$$C_k \leq \rho^{-k}, \quad C_k \leq \frac{6^k}{4}. \quad (4.6)$$

Proof. Applying the coefficient form of Lagrange inversion to $S = t(1 - S)/(1 - 2S)$ gives exactly (4.4). The branch with $S(0) = 0$ of $2S^2 - (1 + t)S + t = 0$ is (4.5). Its first positive branch point is ρ . Since all coefficients are nonnegative, $C_k \rho^k \leq S(\rho) = (1 + \rho)/4 < 1$; convergence at ρ follows also by monotone passage in (4.5). Similarly $C_k 6^{-k} \leq S(1/6) = 1/4$. $\square$

These are the small Schröder coefficients, but no enumerative interpretation is required for any subsequent argument. Retaining the uncanceled mass majorant, rather than the actual total variation after coincident atoms cancel, makes every estimate transparent and reproducible.

Proposition 4.2 (All-slope analytic inverse). *For every nonnegative slope sequence, (4.3) converges normally in a sufficiently small open left half-disc and uniformly on smaller closed left half-discs. More explicitly, it is absolutely convergent whenever*

$$\operatorname{Re} u \leq 0, \quad |u|e^{2\lambda_1|u|} < \rho. \quad (4.7)$$

The resulting function satisfies (1.1) for sufficiently small u in that half-disc. It has $\hat{Q}$ as a uniform Poincaré expansion on each smaller closed left half-disc, as $u \rightarrow 0$.

Proof. Nonnegativity of the slopes gives $b \geq -2k\lambda_1$ at every atom. For $\operatorname{Re} u \leq 0$,

$$\left| u^k \int e^{bu} d\mu_k(b) \right| \leq C_k (|u|e^{2\lambda_1|u|})^k.$$

This proves normal and boundary-uniform convergence.

For a fixed u with negative real part, $q \mapsto \Phi(q, u)$ is holomorphic for $|q| < 1$ and has nonzero derivative $e^{\lambda_1 u}$ at zero. Its local inverse in w has the Lagrange coefficients used above. The same majorant shows convergence of that inverse for small w , uniformly for small u in the left half-disc. For $w = u$ both the inverse series and its value remain sufficiently small; substitution in the convergent q -series gives (1.1). This justifies the functional identity, not only the formal coefficients.

For each fixed N , the blocks with $k < N$ are finite sums of entire exponentials and can be Taylor expanded with a remainder of order u^N . The blocks with $k \geq N$ have a geometric $O(|u|^N)$ tail on smaller half-discs. The constants may depend arbitrarily on N and the slopes. This proves uniform Poincaré asymptotics, without a Gevrey assertion. $\square$

5 Bessel kernels and a ray transform for arbitrary slopes

5.1 One block under the Borel transform

For $k \geq 1$ and $b \in \mathbb{R}$ define the entire function

$$K_{k,b}(\zeta) = \sum_{m \geq 0} \frac{b^m \zeta^{m+k}}{m!(m+k)!} = \frac{\zeta^k}{k!} {}_0F_1(k+1; b\zeta). \quad (5.1)$$

It is the integrated Borel transform of $u^k e^{bu}$. A normalized form of the classical Poisson–Bessel integral [4, Eq. 10.9.4] is particularly effective here.

Lemma 5.1 (Probability-integral representation). *Let*

$$p_k(v) = \frac{\Gamma(k+1)}{\sqrt{\pi}\Gamma(k+1/2)}(1-v^2)^{k-1/2}, \quad -1 < v < 1.$$

Then $\int_{-1}^1 p_k(v) dv = 1$ and

$$K_{k,b}(\zeta) = \frac{\zeta^k}{k!} \int_{-1}^1 \exp(2\sqrt{b}\zeta v) p_k(v) dv. \quad (5.2)$$

The right side is independent of the choice of square root. If $b \geq 0$,

$$|K_{k,b}(\zeta)| \leq \frac{|\zeta|^k}{k!} \exp(2\sqrt{b} |\operatorname{Re} \sqrt{\zeta}|). \quad (5.3)$$

If $b < 0$, replace $\sqrt{b} |\operatorname{Re} \sqrt{\zeta}|$ by $\sqrt{|b|} |\operatorname{Im} \sqrt{\zeta}|$ in that bound. In particular, for $s \geq 0$,

$$|K_{k,b}(-s)| \leq s^k/k!, \quad b \geq 0, \quad (5.4)$$

$$|K_{k,b}(-s)| \leq (s^k/k!)e^{2\sqrt{|b|}s}, \quad b < 0. \quad (5.5)$$

For $b > 0$ there is also the explicit oscillatory formula

$$K_{k,b}(-s) = (-1)^k (s/b)^{k/2} J_k(2\sqrt{bs}). \quad (5.6)$$

Proof. A beta integral gives the normalization and the even moments

$$\int_{-1}^1 v^{2m} p_k(v) dv = \frac{(2m)!k!}{4^m m!(m+k)!}.$$

Odd moments vanish. Expanding the exponential in (5.2) and using these moments gives (5.1). The same evenness proves square-root independence. Taking absolute values under a probability integral proves (5.3) and its negative- b counterpart. For a negative argument and positive b , the integrand is oscillatory and has modulus one. Formula (5.6) follows by comparing the power series for J_k . $\square$

5.2 Exact Laplace recovery and the distinction from a Borel germ

Lemma 5.2 (One-block recovery). *For every $r > 0$, $k \geq 1$, and real b ,*

$$\frac{1}{r} \int_0^\infty e^{-s/r} K_{k,b}(-s) ds = (-r)^k e^{-br}. \quad (5.7)$$

Moreover, the same integral with an absolute value inside is at most r^k for $b \geq 0$, and equals $r^k e^{|b|r}$ for $b < 0$.

Proof. For one fixed b , termwise integration of (5.1) is justified absolutely: the integrated series of absolute values is $r^k \sum_{m \geq 0} (|b|r)^m/m!$. The signed sum is the right side of (5.7). For $b \geq 0$, use the stronger oscillatory bound (5.4) before integrating. For $b < 0$, every term of $(-1)^k K_{k,b}(-s)$ is nonnegative, so the absolute integral is the positive exponential sum just computed. $\square$

Proposition 5.3 (Continuous ray representation without a growth hypothesis). *For arbitrary nonnegative slopes, the function*

$$\mathfrak{B}_Q(-s) = \sum_{k \geq 1} \int K_{k,b}(-s) d\mu_k(b), \quad s \geq 0, \quad (5.8)$$

is well-defined, continuous, and at most exponentially growing in s . For all sufficiently small $r > 0$ its Laplace integral recovers the analytic inverse of Proposition 4.2:

$$Q(-r) = \frac{1}{r} \int_0^\infty e^{-s/r} \mathfrak{B}_Q(-s) ds. \quad (5.9)$$

This alone is not a fine Borel-summability assertion: without a slope bound, (5.8) need not be the restriction of a holomorphic Borel germ at zero.

Proof. Since a negative frequency has $|b| \leq 2k\lambda_1$, the k th term is bounded on the ray by

$$C_k \frac{s^k}{k!} \exp(2\sqrt{2\lambda_1 k s}).$$

For any fixed $\eta > 0$, Young's inequality bounds the last exponential by $\exp(\eta k + 2\lambda_1 s/\eta)$. Sum with $C_k \leq \rho^{-k}$. The result is an exponential in s , and the same bound gives uniform convergence on compact ray intervals.

Lemma 5.2 bounds the integrated absolute contribution of block k by $C_k r^k e^{2k\lambda_1 r}$. Its sum is finite under (4.7). Fubini's theorem is therefore applicable to the kernel sum and Laplace integral; it gives (4.3) at $u = -r$ exactly. No exchange of the original, generally divergent, forward power series with a Laplace integral has been made. $\square$

The last warning is substantive. A continuous ray function with a formal expansion can be produced even when that formal expansion has zero Borel radius. Fine summability requires the additional complex analytic neighborhood proved next.

6 The parabolic continuation and fine growth estimate

6.1 From a slope bound to a block-frequency bound

Lemma 6.1 (Concentration of the maximal frequency). *If $A < \infty$ in (2.5), then for every $a > A$ there is $D \geq 0$ such that every positive frequency in block $k \geq 2$ satisfies*

$$b \leq a(k \log k)^2 + D(k-1). \quad (6.1)$$

Every negative frequency satisfies $|b| \leq 2k\lambda_1$.

Proof. Choose D so that, for all $j \geq 2$, $\lambda_j \leq a(j \log j)^2 + D(j-1)$. Such a D exists by the definition of the limsup and the finiteness of each exceptional slope. The function

$$g(x) = ((x+1) \log(x+1))^2, \quad x \geq 0,$$

is convex and has $g(0) = 0$. Indeed, for $t \geq 1$ the second derivative of $t^2 \log^2 t$ is $2 \log^2 t + 6 \log t + 2 > 0$. Convexity and $g(0) = 0$ imply $g(x) + g(y) \leq g(x+y)$ for nonnegative x, y : bound each term by its fraction of the chord from zero to $x+y$. Therefore

$$\sum_{i=1}^\ell \lambda_{m_i+1} \leq a g\left(\sum_i m_i\right) + D \sum_i m_i = a(k \log k)^2 + D(k-1).$$

Dropping the nonpositive subtraction in (4.2) proves (6.1). The negative-frequency assertion was already used in Proposition 4.2. $\square$

6.2 Normal convergence in the Borel plane

Define

$$B_Q(\zeta) = \sum_{k \geq 1} \int K_{k,b}(\zeta) d\mu_k(b). \quad (6.2)$$

The square-root quantities

$$h(\zeta) = |\operatorname{Re} \sqrt{\zeta}|, \quad \tilde{h}(\zeta) = |\operatorname{Im} \sqrt{\zeta}|$$

are independent of the sign of the root, and $h(\zeta)^2 = (|\zeta| + \operatorname{Re} \zeta)/2$. By Lemmas 5.1 and 6.1, the absolute mass of block $k \geq 2$ is bounded by

$$\frac{\rho^{-k} |\zeta|^k}{k!} \exp\left(2\sqrt{a} h(\zeta) k \log k + 2\sqrt{D} h(\zeta) \sqrt{k} + 2\sqrt{2\lambda_1} \tilde{h}(\zeta) \sqrt{k}\right). \quad (6.3)$$

The estimate combines the positive- and negative-frequency bounds; an individual atom needs only one of the corresponding exponents.

Let K be compact and suppose $\sup_K 2\sqrt{a} h < 1$. Taking logarithms in (6.3) and using Stirling's elementary estimate shows that the k th bound is

$$\exp(-(1 - \beta)k \log k + O_K(k) + O_K(\sqrt{k})), \quad \beta = \sup_K 2\sqrt{a} h < 1.$$

This is summable. Thus (6.2) converges normally on $\{2\sqrt{a} h < 1\}$. If $A > 0$, every compact subset of (2.6) admits a choice $a > A$ satisfying this strict inequality. If $A = 0$, every compact subset of $\mathbb{C}$ admits a sufficiently small $a > 0$. We obtain the continuation assertions of Theorem 2.4.

In a neighborhood of zero, normal convergence also permits Taylor coefficient extraction. Each kernel in block k vanishes to order k ; hence the coefficient of ζ^n is a finite sum and is exactly $v_n/n!$. Consequently (6.2) really is the analytic continuation of $\mathcal{B}_0 \hat{Q}$, not merely a ray function chosen to have a desired Laplace transform.

6.3 Exponential bounds in a fixed strip

Parabolic continuation alone does not prove fine summability; growth must be controlled on a fixed-width neighborhood of the entire ray. We give that estimate explicitly.

Choose one $a > A$ and choose $\delta > 0$ small enough that the closure of $\Omega_{2\delta}^-$ stays inside $\{2\sqrt{a} h < 1\}$. This is possible because $h^2 \leq 2\delta$ on that tube. For $\zeta = -s + it$, $s \geq 1$, $|t| \leq \delta$, one has

$$h(\zeta) \leq \frac{\delta}{2\sqrt{s}}, \quad \tilde{h}(\zeta) \leq \sqrt{s + \delta}, \quad |\zeta| \leq s + \delta. \quad (6.4)$$

The first estimate follows by rationalizing $\sqrt{s^2 + t^2} - s$.

Absorbing fixed constants into d_1, d_2, d_3 , the logarithm of the k th bound in (6.3) is at most

$$k \log(\rho^{-1}(s + \delta)) - \log k! + \frac{d_1}{\sqrt{s}} k \log k + \frac{d_2}{\sqrt{s}} \sqrt{k} + d_3 \sqrt{ks}. \quad (6.5)$$

Split the sum into $k \leq s^2$ and $k > s^2$.

In the first range, $(d_1/\sqrt{s}) \log k \leq 2d_1 \log s/\sqrt{s} = o(1)$. Young's inequality bounds the last two terms by $c_1 k + c_2 s + c_3$. For large s , the first part of the sum is consequently at most

$$C e^{c_2 s} \sum_{k \geq 1} \frac{(\rho^{-1} e^{c_1 + 1} (s + \delta))^k}{k!} \leq C' e^{H_1 s}.$$

In the second range, $\log(s + \delta) \leq \frac{1}{2} \log k + o(1)$. For sufficiently large s we have $d_1/\sqrt{s} \leq 1/8$. Using $\log k! \geq k \log k - k$ in (6.5), while $\sqrt{ks}/k \leq 1/\sqrt{s}$, gives an upper bound $-\frac{1}{4}k \log k$ after increasing the threshold on s . The tail is thus uniformly bounded. The remaining bounded portion of the tube is compactly contained in the domain of normal convergence. We conclude that

$$|B_Q(\zeta)| \leq Ce^{H|\zeta|}, \quad \zeta \in \Omega_\delta^-. \quad (6.6)$$

This proves the remaining assertion of Theorem 2.4 and, in particular, the implication (i) $\Rightarrow$ (v) of Theorem 2.3. Proposition 5.3 then identifies the Borel sum with the literal analytic inverse Q .

7 Summation commutes with taking the inverse

The preceding construction proves fine summability of $\widehat{Q}$ directly. To transfer the result to $\widehat{U}$, one needs closure under compositional inversion, not just closure of Gevrey coefficient classes. This is a classical nonlinear summation principle. We include a proof adapted to fixed half-strips, following the convolution method underlying [5, Sections 13 and 17]. This avoids silently replacing a half-strip hypothesis by an open angular-sector hypothesis.

7.1 The elementary half-strip algebra

At infinity write $\varphi(z) = \sum_{n \geq 1} \varphi_n z^{-n}$ and use its Borel minor

$$\mathcal{B}\varphi(\xi) = \sum_{n \geq 1} \frac{\varphi_n}{(n-1)!} \xi^{n-1}.$$

Let $\Omega_\delta^+ = -\Omega_\delta^-$. Consider those minors holomorphic there and bounded by $Me^{c|\xi|}$. The tube is convex, so every straight segment from zero to a point of the tube stays inside it. If $\widehat{\varphi}$ is such a minor, its k -fold convolution obeys

$$|\widehat{\varphi}^{*k}(\xi)| \leq M^k \frac{|\xi|^{k-1}}{(k-1)!} e^{c|\xi|}. \quad (7.1)$$

Indeed, parameterize the convolution simplex by nonnegative coordinates summing to one; the exponential factors multiply to $e^{c|\xi|}$ and the simplex volume is $1/(k-1)!$.

Products of formal series correspond to convolution, with the constant terms handled separately. Formula (7.1) also shows that substitution into a convergent scalar power series is permitted whenever the substituted series has zero constant term. If $H(w) = \sum h_k w^k$ and $|h_k| \leq CR^{-k}$, the sum of the Borel minors of $\sum_{k \geq 1} h_k \varphi^k$ is bounded by a constant times $\exp((c + M/R)|\xi|)$. In particular, reciprocals of series with nonzero constant term belong to the same class.

Multiplication by z , with the new polynomial part separated off, corresponds to differentiation of the minor. Cauchy's estimate on a narrower tube preserves an exponential bound. Division by z corresponds to integration. Translation $z \mapsto z + c_0$ multiplies the minor by $e^{-c_0 \xi}$ and preserves the class. Laplace summation commutes with all these operations: products follow from Fubini on convolution integrals, reciprocal and scalar substitution from the same absolutely convergent convolution majorant, multiplication by z from integration by parts, and translation from its exponential multiplier. All these identities hold on a sufficiently far right half-plane.

7.2 A direct inverse formula and its analytic meaning

Lemma 7.1 (Fine-summable inverse at infinity). *Suppose $f(z) = z + c_0 + \varphi(z)$, where $\varphi \in z^{-1}\mathbb{C}[[z^{-1}]]$ has a fine-summable minor in the positive direction. Then the formal inverse*

$g(z) = z - c_0 + O(z^{-1})$ has the same property. Its sum is the analytic inverse of the sum of f on sufficiently far right half-planes.

Proof. First take $c_0 = 0$. Formal Lagrange inversion gives

$$g(z) = z + \chi(z), \quad \chi = \sum_{k \geq 1} \frac{(-1)^k}{k!} \partial_z^{k-1}(\varphi^k). \quad (7.2)$$

One way to verify this formula is to solve $\xi = -t\varphi(z + \xi)$ in the formal parameter t and then put $t = 1$; the order in z^{-1} tends to infinity with k . Since differentiation corresponds to multiplication by $-\xi$ in the Borel plane, the candidate minor is

$$\hat{\chi}(\xi) = - \sum_{k \geq 1} \frac{\xi^{k-1}}{k!} \hat{\varphi}^{*k}(\xi). \quad (7.3)$$

By (7.1), its absolute value is at most

$$e^{c|\xi|} \sum_{k \geq 1} \frac{M^k |\xi|^{2k-2}}{k!(k-1)!} \leq M \exp((c + 2\sqrt{M})|\xi|).$$

The last inequality follows by comparison with $\sum_{j \geq 0} (\sqrt{M}|\xi|)^{2j}/(j!)^2 \leq \exp(2\sqrt{M}|\xi|)$. Normal convergence proves the required holomorphy on the same half-strip.

It remains to identify the sum as an inverse function. The Laplace sum $\varphi_*(z)$ satisfies $|\varphi_*(z)| \leq M/(\operatorname{Re} z - c)$ and $|\varphi'_*(z)| \leq M/(\operatorname{Re} z - c)^2$. For $\operatorname{Re} z$ sufficiently large and $|t| \leq 2$, the equation $\xi = -t\varphi_*(z + \xi)$ is a contraction on $|\xi| \leq 1$. Its solution is holomorphic in t on a disc of radius greater than one. The ordinary analytic Lagrange formula is therefore convergent at $t = 1$ and gives (7.2) with φ_* in place of φ . On the other hand, the exponential majorant for (7.3) permits termwise Laplace integration, and each term sums to the corresponding term of that analytic Lagrange formula. Thus the sum of g really is the inverse of $z + \varphi_*(z)$.

For a nonzero constant, write $f = \tau_{c_0} \circ f_0$, where $f_0 = z + \varphi$ and $\tau_{c_0}(z) = z + c_0$. Then $f^{(-1)} = f_0^{(-1)} \circ \tau_{-c_0}$. Translation, already proved to preserve fine summability and its sums, completes the argument. $\square$

7.3 Return to the feedback variables

For a tangent-to-identity series Q , conjugate by the involution $J(z) = -1/z$:

$$f = J \circ Q \circ J, \quad f(z) = -\frac{1}{Q(-1/z)} = z + c_0 + O(z^{-1}). \quad (7.4)$$

Fine summability of Q on the negative ray gives the positive-direction half-strip minor of $Q(-1/z)$ by reflection and differentiation of $B_Q(-\xi)$. The algebra operations just proved, including multiplication by z and reciprocal of a nonzero constant term, give the fine-summable tail of f . Lemma 7.1 gives $g = f^{(-1)}$, and conjugating back gives $J \circ g \circ J = Q^{(-1)} = U$. Each operation respects the summed functions. We have proved fine summability of U , and that the two sums are inverses, without assuming a larger angular continuation domain.

Completion of Theorem 2.3. The slope bound implies fine summability of Q by Theorem 2.4, and the preceding argument implies fine summability of U . Fine summability implies Gevrey-1. The two Gevrey conditions are equivalent by Lemma 3.2, and either implies the slope condition by Lemma 3.1. This proves all five equivalences.

The sum of Q is the literal inverse constructed in Proposition 4.2, by exact Laplace recovery. For small negative q , its inverse satisfies $U(q) = q + O(q^2) < 0$. Putting $u = U(q)$ in (1.1) gives

(2.4); its absolute convergence follows from $\operatorname{Re} u < 0$ and $|q| < 1$. On a sufficiently small proper left sector, $U(q) = q + O(q^2)$ similarly implies $\operatorname{Re} U(q) < 0$, proving the stated complex version on the common domain. $\square$

8 Quadratic feedback: exact blocks and uniform remainders

8.1 A normalized expansion with nonnegative frequencies

Take $\lambda_j = j^2$ and $P = e^u Q$. In an atom from (4.2), the new frequency is

$$a_{k,\mathbf{m}} = b_{k,\mathbf{m}} + 1 = \sum_{i=1}^{\ell} m_i(m_i + 1). \quad (8.1)$$

Because the parts sum to $k - 1$,

$$0 \leq a_{k,\mathbf{m}} \leq k(k - 1). \quad (8.2)$$

The upper bound follows from $\sum m_i^2 \leq (\sum m_i)^2$. Thus there are explicitly computable polynomials $H_k(v)$ with

$$P(u) = \sum_{k \geq 1} u^k H_k(e^u), \quad \sum_a |[v^a] H_k(v)| \leq C_k, \quad \deg H_k \leq k(k - 1). \quad (8.3)$$

The first four are

$$\begin{aligned} H_1(v) &= 1, & H_2(v) &= -v^2, \\ H_3(v) &= 2v^4 - v^6, & H_4(v) &= -5v^6 + 5v^8 - v^{12}. \end{aligned}$$

They are not the ordinary Taylor coefficients a_n of P ; the two levels of expansion must be kept distinct.

Since all frequencies in (8.3) are nonnegative, this block expansion converges uniformly on every closed left half-disc $|u| \leq r_0 < \rho$. A directly usable tail bound, for $r < 1/6$, is

$$\left| P(u) - \sum_{k=1}^K u^k H_k(e^u) \right| \leq \frac{(6r)^{K+1}}{4(1 - 6r)}, \quad \operatorname{Re} u \leq 0, \quad |u| \leq r. \quad (8.4)$$

For Q , multiply this bound by e^r . This is a convergent block approximation to a function with a divergent ordinary Taylor expansion. There is no contradiction: each block retains an entire exponential instead of Taylor expanding it indefinitely.

8.2 Proof of the uniform remainder certificate

For $m \geq 1$, the integral form of Taylor's remainder says

$$e^z - \sum_{j=0}^{m-1} \frac{z^j}{j!} = \frac{z^m}{(m-1)!} \int_0^1 (1-t)^{m-1} e^{tz} dt.$$

If $\operatorname{Re} z \leq 0$, its modulus is at most $|z|^m/m!$. Apply this to each exponential in block $k < N$, using $m = N - k$ and (8.2). The total error from this block is at most

$$r^N C_k \frac{[k(k-1)]^{N-k}}{(N-k)!}.$$

The unexpanded blocks $k \geq N$ contribute at most

$$\sum_{k \geq N} \rho^{-k} r^k \leq r^N \frac{\rho^{-N}}{1 - r_0/\rho}.$$

The finite Taylor sums assemble to exactly $\sum_{n < N} a_n u^n$, because the coefficient at degree n uses only $k \leq n$. Summing the bounds proves (2.8) and (2.9). Notice that this estimate is uniform all the way to the imaginary boundary of the half-disc, where ordinary exponential damping disappears.

9 Asymptotics of the majorant and the deeper flat scale

9.1 A discrete saddle for the positive majorant

We now prove (2.10), including its linear constant. For $2 \leq k < N$ define

$$T_{N,k} = C_k \frac{[k(k-1)]^{N-k}}{(N-k)!}.$$

The $k = 1$ summand vanishes. The logarithm of a sum of at most N positive terms differs from its largest logarithm by at most $\log N$. The geometric-tail term in (2.9) has logarithm only $O(N)$. It will be smaller than the saddle contribution.

Put $t = k/N$ and $c = \rho^{-1}$. The bounds $1 \leq C_k \leq c^k$, $k(k-1) \leq k^2$, and the uniform elementary Stirling estimate give

$$\frac{1}{N} \log T_{N,k} \leq (1-t)(\log N + 2 \log t - \log(1-t) + 1) + t \log c + O(\log N/N). \quad (9.1)$$

The error bound is uniform for $1 \leq N-k \leq N$. Set $a = t \log N$. Subtracting $\log N - 2 \log \log N$ from the right side of (9.1) gives

$$2 \log a + 1 - a + R(t) + O(\log N/N), \quad (9.2)$$

where

$$R(t) = -2t \log t - (1-t) \log(1-t) - t + t \log c.$$

The function R is bounded on $[0, 1]$, with its endpoint values interpreted continuously, and $R(t) \rightarrow 0$ as $t \rightarrow 0$.

The function $2 \log a + 1 - a$ tends to minus infinity both as $a \downarrow 0$ and as $a \rightarrow \infty$. Consequently any indices capable of maximizing (9.2) must have a in a fixed compact subinterval of $(0, \infty)$. On such a set, $t = a/\log N \rightarrow 0$ uniformly, and hence $R(t) = o(1)$. The limiting optimization is therefore exact:

$$\max_{a>0} (2 \log a + 1 - a) = 2 \log 2 - 1 = \log 4 - 1, \quad a = 2.$$

For a matching lower bound choose $k = \lfloor 2N/\log N \rfloor$. Use $C_k \geq 1$ and Stirling's formula. Replacing $k(k-1)$ by k^2 changes the logarithm by $(N-k) \log(1-1/k) = O(\log N) = o(N)$. The same computation without $t \log c$ therefore attains $\log 4 - 1 + o(1)$. The maximum and then the positive sum have precisely the asymptotic (2.10). This completes the proof of Theorem 2.5.

Corollary 9.1 (Uniform Gevrey-1 of zero type). *For every $H > 0$ and every fixed $r_0 < \rho$ there is a constant C_H such that the remainder in (2.8) is at most $C_H H^N N! |u|^N$ for all $N \geq 2$ and all u in the stated half-disc.*

Proof. From (2.10) and Stirling,

$$\frac{1}{N} \log \frac{M_N}{N!} = -2 \log \log N + \log 4 + o(1) \rightarrow -\infty.$$

Thus $M_N \leq H^N N!$ for sufficiently large N ; enlarge C_H for the finitely many remaining indices. $\square$

9.2 Optimizing the proved bound

Proof of Theorem 2.6. Let $L = \log(1/r)$ and $N = \lfloor L^2/(4r) \rfloor$. Then

$$\log N = L + 2 \log L - \log 4 + o(1), \quad \log \log N = \log L + o(1).$$

Using (2.10),

$$\begin{aligned} \log(r^N M_N) &= N(-L + \log N - 2 \log \log N + \log 4 - 1 + o(1)) \\ &= -\frac{L^2}{4r}(1 + o(1)). \end{aligned}$$

The finite bound (2.8) is uniform over the half-circle, so this proves (2.11). Multiplication by e^r contributes only r to the logarithm of the error bound, which is absorbed by an arbitrarily small loss in ε . $\square$

Under the transseries substitution $u = -e^{-X}$, $X \rightarrow +\infty$, the remainder scale becomes

$$\exp\left[-\left(\frac{1}{4} - o(1)\right) X^2 e^X\right]. \quad (9.3)$$

This is invisible to every fixed exponential monomial e^{-nX} . It is an analytically proved error scale arising from an infinite reorganization of a one-generator formal exponential transseries. The conclusion is an upper bound; it does not by itself specify a new nonzero transseries sector or prove that this scale appears with a nonvanishing coefficient in a completed expansion.

9.3 What the constant does and does not say

There are three different quantities here: the growth of the actual Taylor coefficients, the growth of the positive majorant M_N , and the smallest actual error among all truncations. The saddle calculation proves a sharp asymptotic for the second quantity. It proves an upper bound for the third, but no matching lower bound. The signed block coefficients can cancel, so replacing “majorant asymptotic” by “actual optimal remainder asymptotic” would be unjustified.

Likewise, an entire Borel transform need not have acceptable growth in every direction. The earlier feedback draft establishes, for the forward quadratic model, an entire ordinary Borel transform with excessive growth on the positive ray [2]. Our negative-ray recovery does not contradict that obstruction. It supplies the missing favorable direction, and it does not assert an angular neighborhood of favorable directions.

For finite computation, use (2.9) rather than the limiting constant. The convergence in (2.10) is slow: the packaged floating-point diagnostics still place the centered value $(\log M_N - N \log N + 2N \log \log N)/N$ near 1.35 at $N = 10^5$, whereas its proved limit is $\log 4 - 1 \approx 0.38629$. This is a reason to retain explicit certificates, not a discrepancy with a limit containing an unspecified $o(N)$ term.

10 Examples, exact computations, and inverse certificates

10.1 Examples across the threshold

For $\lambda_j = j^2(\log(j+1))^\beta$, the conclusion is particularly simple:

Slope regime	Fine negative summation	Guaranteed inverse Borel domain
$\beta < 2$	Yes	Entire plane
$\beta = 2$	Yes	$ \zeta + \operatorname{Re} \zeta < 1/2$
$\beta > 2$	No	No holomorphic Borel germ at zero

The failure in the last row holds for both $\widehat{U}$ and $\widehat{Q}$. Nevertheless, Propositions 4.2 and 5.3 still construct the literal negative-side inverse and its continuous ray representation. The criterion is a limsup criterion: slopes exceeding the threshold by unbounded factors on an arbitrarily sparse subsequence already prevent fine Borel summability. Eventual monotonicity or regular variation of the slopes is not assumed.

For comparison, the convergence threshold is much smaller: $\widehat{U}$ and $\widehat{Q}$ are convergent at zero exactly when $\lambda_j = O(j)$, as already recorded in [2]. Here is a short independent check. If $\lambda_j \leq Lj$, the kernel is holomorphic for $|q|e^{L|u|} < 1$ and the analytic implicit-function theorem applies at $(0, 0)$. Conversely, if $u_n \leq CH^n$, use (3.2) with $m = j$ to obtain $\lambda_j \leq C^{1/j}H^2(j!)^{1/j} = O(j)$. Convergent reversion preserves analyticity. Thus in the quadratic model both original Taylor series diverge, even though the inverse Borel transform constructed here is entire.

Two soluble cases test the normalizations. With all slopes zero, $U(q) = q/(1 - q)$ and $Q(u) = u/(1 + u)$. With $\lambda_j = j$,

$$Q(u) = \frac{ue^{-u}}{1 + u}. \quad (10.1)$$

The verification program checks these against a separate forward coefficient recurrence.

10.2 Independent exact coefficient checks

The accompanying program computes the forward coefficients from $E_j(q) = \exp(j^2 U(q))$ using

$$[q^m]E_j = \frac{j^2}{m} \sum_{i=1}^m i u_i [q^{m-i}]E_j, \quad u_n = \sum_{j=1}^n [q^{n-j}]E_j. \quad (10.2)$$

The first formula follows by differentiating E_j ; the second is the feedback equation. This computation is independent of the inverse-block construction. The latter uses dynamic programming over compositions, retaining exact rational coefficients and integer frequencies. The first ordinary coefficients are as follows.

n	u_n in $\widehat{U}$	a_n in $\widehat{P}$	v_n in $\widehat{Q}$	C_n
1	1	1	1	1
2	2	-1	-2	1
3	15/2	-1	1/2	3
4	107/3	-1	-2/3	11
5	1555/8	-13/3	-29/8	45
6	17362/15	-49/3	-743/60	197
7	5288311/720	-1099/15	-8491/144	903
8	15414767/315	-16207/45	-92726/315	4279

The executed exact-arithmetic checks verify both $\widehat{U} \circ \widehat{Q} = u$ and $\widehat{Q} \circ \widehat{U} = q$ through degree 24, the recurrence of S , the block-frequency bounds, and the uncanceled-mass bounds. The zero- and linear-slope tests also pass through degree 24. These are finite checks of formulas and implementation, not substitutes for the preceding proofs.

10.3 Three rational enclosures of the inverse function

The block series gives numerical certificates without first estimating a small divergent-series remainder. For rational $r > 0$, enclose $t = e^{-r}$ between consecutive Taylor polynomials. Taylor's integral remainder gives the direction of the two inequalities exactly. Evaluate the finite polynomial $\sum_{k=1}^K (-r)^k H_k(t)$ by rational interval arithmetic, append the rational tail bound (8.4), and divide by the positive interval for t . This encloses $Q(-r) = P(-r)/t$.

For $K = 14$, the executed program used Taylor degrees 36 and 37 for t , then rounded the final rational endpoints outward to 24 decimal places. Every displayed endpoint below is therefore an exact terminating rational, not a rounded floating-point claim.

r	Lower bound for $Q(-r)$	Upper bound for $Q(-r)$
1/20	-0.055065695744769945932257	-0.055065684970061950441109
1/40	-0.026258040219395119101712	-0.026258040219131013674068
1/100	-0.010200506315988279498804	-0.010200506315988279246191

The corresponding widths are less than 1.078×10^{-8} , 2.642×10^{-13} , and 2.527×10^{-19} , respectively; the unrounded exact widths and all certificate inputs are in the data files.

10.4 A certified derivative bound and error transport

Proposition 10.1 (A concrete inverse-error certificate). *For quadratic feedback and $u \in [-1/20, 0]$, the inverse branch obeys (the derivative at zero is interpreted from the left)*

$$|Q(u)| < 1/10, \quad Q'(u) \geq \frac{619}{900} > 0. \quad (10.3)$$

If u_* and v lie in this interval and $Q(u_*) = q_*$, then

$$|u_* - v| \leq \frac{900}{619} |Q(v) - q_*|. \quad (10.4)$$

If $Q(v)$ is known only through an approximation $\tilde{Q}(v)$ with certified error η , its residual in (10.4) can be replaced by $|\tilde{Q}(v) - q_*| + \eta$. Existence of u_* must first be checked by a bracket; an error bound alone is not an existence proof.

Proof. For $r \leq 1/20$, (4.6) gives

$$|Q(u)| \leq e^r S(r) \leq \frac{1}{1-r} \left(r + \frac{9r^2}{1-6r} \right) < \frac{1}{10}.$$

The same estimate holds on the closed left half-disc. It guarantees that (1.1), initially proved near zero, continues throughout its open half-disc by the identity theorem, since the q -series stays inside $|q| < 1$.

Let $R = 1/10$. On the indicated real interval the differentiated kernel satisfies

$$|\Phi_u(Q(u), u)| \leq \sum_{j \geq 1} j^2 R^j = \frac{R(1+R)}{(1-R)^3} = \frac{110}{729},$$

and

$$\Phi_q(Q(u), u) \geq e^u - \sum_{j \geq 2} j R^{j-1} \geq \frac{19}{20} - \frac{19}{81} > 0, \quad \Phi_q(Q(u), u) \leq \frac{100}{81}.$$

These derivatives are justified by absolute convergence of their geometric majorants. Differentiating $u = \Phi(Q(u), u)$ yields

$$Q'(u) = \frac{1 - \Phi_u}{\Phi_q} \geq \left(1 - \frac{110}{729} \right) \frac{81}{100} = \frac{619}{900}.$$

The mean-value theorem proves (10.4); the approximate residual follows from the triangle inequality. $\square$

As an executed application, take $q_* = -1/40$. The first rational enclosure above and $Q(0) = 0$ bracket this target, so a unique $u_* = U(-1/40)$ lies in $[-1/20, 0]$. Choose the rational center $v = -0.02385490927750544$, without assuming its accuracy. Enclosing $Q(v)$ by 14 blocks and Taylor degrees 12 and 13 for e^v , then applying (10.4) and outward rounding, gives the exact certificate

$$\boxed{-0.023854909277599602576607 \leq U(-1/40) \leq -0.023854909277411277423393.} \quad (10.5)$$

The width is $1.88325153214 \times 10^{-13}$. The center was selected numerically, but its certificate uses only rational arithmetic and Proposition 10.1.

10.5 Floating-point checks are separately labeled

At 70 decimal digits, the program compares the 24-block inverse approximation with a numerical root of a 179-term literal feedback sum at $r = 0.01, 0.025, 0.05$. It also numerically integrates four independent Bessel kernels and compares the integrals with (5.7). All four kernel comparisons have absolute discrepancy below 10^{-60} in that run. These calculations are floating-point diagnostics, not interval proofs of the roots or of the integrals. The rational enclosures and residual-transport certificate are separate computations and do not depend on those comparisons.

The majorant diagnostic program evaluates a positive sum in log-space and records the slow movement of its maximizing block. It uses the Schröder recurrence

$$kC_k = 3(2k-3)C_{k-1} - (k-3)C_{k-2}, \quad k \geq 3,$$

in ratio form. Its output is explicitly marked non-certified; the proof of the limiting saddle does not use that output.

11 Further research questions and proposed directions

The following questions are not claimed solved in this article. They are framed so that a successful answer would add information not already contained in the fine summation theorem.

11.1 Angular summability of the quadratic model

Determine the set of angles θ for which the entire functions $\mathcal{B}_0\widehat{Q}(te^{i\theta})$ and $\mathcal{B}_0\widehat{U}(te^{i\theta})$ have at most exponential growth as $t \rightarrow \infty$, with the uniform neighborhood needed for angular summability. In particular, does any open arc containing π work? The present fixed-width tube does not answer that question. The uncanceled Bessel estimate deteriorates off the negative ray, so a proof of an open arc would have to exploit cancellation or a different exact representation. A proof that no such arc exists would identify a genuine boundary between fine and angular summability in a concrete nonlinear implicit model.

11.2 The actual Borel boundary at critical slope growth

For $\lambda_j \sim a(j \log j)^2$, $a > 0$, determine the singularities of $\mathcal{B}_0\widehat{Q}$ and whether any part of the boundary $|\zeta| + \operatorname{Re} \zeta = 1/(2a)$ is an actual obstruction. Our theorem proves normal convergence strictly inside; it does not prove failure outside. A related question is the exact factorial-normalized type of $\widehat{Q}$ and its relation to the forward type. Class membership is invariant under reversion, but the elementary proof here does not establish a numerical type identity. This is a useful place for a more detailed literature comparison before asserting novelty of any type-preservation theorem.

11.3 Signed coefficient asymptotics and actual least error

Find an asymptotic equivalent, including signs and prefactors, for the ordinary coefficients of $\hat{P}$ or $\hat{Q}$ in the quadratic case. The positive majorant deliberately ignores cancellation between compositions with different lengths. A signed saddle analysis may reveal a smaller coefficient scale or a different correction to the dominant block. The stronger analytic target is a matching lower bound for the best Taylor-truncation error, or a full description of its exceptional zeros. Only then would it be legitimate to interpret a constant such as $1/4$ as the actual least-error action rather than a certificate constant.

11.4 Other powers and generalized summation kernels

For regularly varying slopes $\lambda_j \sim aj^p$ with $p > 1$, construct analogues of the inverse kernel formula for moment Borel transforms with weights $\Gamma(1 + (p - 1)n)$. Determine which directions have sufficient kernel damping, and whether the literal left-side inverse is recovered. For $p > 2$ the ordinary Borel germ need not exist, so changing the summation order is essential rather than cosmetic. The quadratic proof suggests that keeping exponential blocks intact may remain useful, but it does not establish a generalized summability theorem for those other orders.

11.5 Complex slopes, signs, and loss of positivity

Replace the nonnegative slopes by complex slopes, perhaps contained in a fixed sector, and identify sufficient geometric conditions for an oscillatory or damped Borel direction. Then seek necessity statements. Our forward lower bound uses positivity and generally fails when coefficients cancel; the negative-ray Bessel bound also uses the slopes' common real direction. Both parts of the equivalence must therefore be revisited. A common half-plane of damping for the literal kernel may suffice for analytic inversion without sufficing for fine summability.

11.6 Weighted actions and joint amplitude–slope thresholds

Study

$$U(q) = \sum_{j \geq 1} c_j q^j e^{\lambda_j U(q)}, \quad c_1 = 1,$$

with positive amplitudes that can decay rapidly. Determine the sharp joint condition on (c_j, λ_j) for fine summability and for convergence. The inverse-block formula survives with additional products of the c_j , but the mass generating function and the positive coefficient lower bound both change. Very small amplitudes can suppress isolated large slopes, so a criterion involving only $\sup \lambda_j/j$ or a bare slope limsup should not be expected in this setting.

11.7 Canonical recovery beyond every finite Gevrey class

For superpolynomial slopes, the all-slope ray construction still produces a continuous, exponentially bounded transform recovering the literal inverse. Relate this transform to acceleration, moment summability, or other summation procedures defined from formal coefficients. The main tasks are to specify a summation class and prove that its canonical value coincides with the literal implicit branch. Calling the continuous ray function a Borel germ would evade, rather than solve, this problem. Uniqueness should be proved within the chosen summation class and not inferred merely from having a common Poincaré expansion.

11.8 Sharper certified algorithms and automatic normalization

Develop an algorithm that recognizes when a normalization such as $P = e^u Q$ removes all negative block frequencies, then selects between geometric block truncation and ordinary Taylor truncation using certified cost and error estimates. Improving the uncanceled mass bound could substantially reduce the number of blocks needed in rational certificates. A further goal is an interval-verified Borel–Laplace quadrature directly from the Bessel representation, with rigorously bounded tails and a verified equivalence to the simpler block evaluator.

12 A formalization roadmap and proof obligations

The repository’s existing formal scales and support infrastructure are relevant context, but they do not automatically formalize any analytic summation assertion proved here. A practical staged development would separate finite algebra from measure, complex-analysis, and uniform-limit arguments.

The first stage should formalize the composition-indexed coefficient measures as finitely supported maps, prove the exact frequency formulas, and establish the mass recurrence and generating-function identities. The quadratic bound $0 \leq a \leq k(k-1)$ and the two composition checks are finite algebraic targets. They require no Borel integration theory. An exact certificate checker can verify the rational interval arithmetic using proved exponential Taylor bounds and the geometric tail theorem.

The second stage should establish the normalized beta moments and (5.2), followed by the one-block Laplace identity. The proof must distinguish an absolutely integrable Taylor series for one fixed frequency from the much stronger absolute-integrability estimate used when summing infinitely many blocks. Confusing those two exchanges would invalidate the all-slope construction.

The third stage should formalize the compact-normal convergence estimate (6.3) and the fixed-strip split at $k = s^2$. Only after those are in place should the implementation identify the formal Borel coefficients of the limiting function. The nonlinear inverse theorem then belongs in a reusable library for half-strip Borel minors and Laplace summation, rather than being hidden in a feedback-specific lemma.

Finally, the logarithmic saddle proof should have an explicit statement about the *positive majorant*, with a separate theorem transporting it to an upper bound on a function error. This distinction is valuable for formalization: it prevents a later user from inadvertently rewriting an inequality as an asymptotic equivalence for an oscillating remainder. No Lean files are supplied with this article, and no claim is made that these stages have been implemented or checked.

13 Conclusion

The countable feedback model provides a concrete example in which formal support, coefficient growth, directional summation, and uniform error control are genuinely different layers. Inverting the dependent variable first preserves enough exponential structure to bridge several of them. The resulting Bessel representation makes the negative direction accessible exactly up to the known Gevrey–1 slope threshold, identifies the summed functions with the literal implicit branch, and yields a stronger quantitative theorem in the quadratic case.

The proven progress is therefore specific: a negative-direction fine summation theorem, a guaranteed inverse Borel continuation region, a closed-half-disc logarithmic-Gevrey certificate, and reproducible exact inverse bounds. Angular summability, maximal continuation, global resurgence, and actual optimal-remainder asymptotics remain separate research questions. These boundaries are part of the result, not qualifications to be removed in a future summary.

A Reproducibility and package contents

The archive contains the self-contained source `article.tex`, the compiled `article.pdf`, the two programs in `code/`, and recorded data in `data/`. The primary exact-arithmetic program uses Python's standard-library fractions. If `mpmath` is installed, it also runs the separately labeled numerical diagnostics. The majorant program uses only the standard library.

The exact commands used to generate the data and document are

```
python code/verify.py -order 24
python code/majorant.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The recorded run used Python 3.13.5 and `mpmath` 1.3.0 at 70 digits for the floating-point checks. The scripts accept ordinary local execution and make no network requests. No source file or branch in ProveIt was modified.

The certificate files store rational numerators and denominators in addition to decimal displays. The two composition identities are checked by truncated polynomial composition, not by comparing a few numerical function values. Re-running at a larger order extends those finite checks but does not change the status of the analytic proofs.

B An explicit coefficient-extraction algorithm

For the quadratic blocks, let $D_{s,\ell}(a)$ count ordered compositions of s into ℓ positive parts whose frequency $\sum m_i(m_i + 1)$ equals a . Start with $D_{0,0}(0) = 1$, and use

$$D_{s,\ell}(a) = \sum_{m=1}^s D_{s-m,\ell-1}(a - m(m+1)). \quad (\text{B.1})$$

Absent indices have value zero. Then

$$[v^a]H_k(v) = \frac{1}{k} \sum_{\ell=1}^{k-1} (-1)^\ell \binom{k+\ell-1}{\ell} D_{k-1,\ell}(a), \quad k \geq 2. \quad (\text{B.2})$$

Finally,

$$a_n = \sum_{k=1}^n \sum_a [v^a]H_k(v) \frac{a^{n-k}}{(n-k)!}, \quad v_n = \sum_{j=0}^{n-1} \frac{(-1)^j}{j!} a_{n-j}. \quad (\text{B.3})$$

The convention $0^0 = 1$ handles $H_1 = 1$. All these expressions are finite and rational. Equation (B.1) aggregates compositions with the same frequency, avoiding explicit enumeration of all ordered compositions. It is a reproducible reference algorithm, not a claim of asymptotically optimal computational complexity.

References

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